Math 121: Commutative Algebra

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Problem 1 (5 points). Let $\varphi : R \to S$ be a ring homomorphism. Carefully prove that the induced map on spectra

$$\varphi^{\#} : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$$

is continuous in the Zariski topology.

Solution.

Define

$$\begin{aligned} \varphi^{\#} &: \operatorname{Spec}(S) \to \operatorname{Spec}(R) \\ \varphi^{\#}(q) &\mapsto \varphi^{-1}(q) \end{aligned}$$

To show that $\varphi^{\#}$ is continuous, we must show that the preimage of a closed set $V \in \operatorname{Spec}(R)$ is closed in $\operatorname{Spec}(S)$.

Closed sets in $\operatorname{Spec}(R)$ are of the form

$$\mathbb{V}(\mathcal{S}) = \{[p] \in \operatorname{Spec}(R) \mid \mathcal{S} \subseteq p\}$$

Consider $(\varphi^{\#})^{-1}(\mathbb{V}(\mathcal{S}))$,

$$\begin{aligned} (\varphi^{\#})^{-1}(\mathbb{V}(\mathcal{S})) &= \{[q] \in \operatorname{Spec}(S) \mid \varphi^{\#}([q]) \in \mathbb{V}(\mathcal{S})\} \\ &= \{[q] \in \operatorname{Spec}(S) \mid \mathcal{S} \subseteq \varphi^{\#}([q])\} \end{aligned}$$

Since the pre-image of a prime ideal is a prime ideal we have,

$$\begin{aligned} (\varphi^{\#})^{-1}(\mathbb{V}(\mathcal{S})) &= \{[q] \in \operatorname{Spec}(S) \mid \varphi(\mathcal{S}) \subseteq [q]\} \\ &= \mathbb{V}(\varphi(\mathcal{S})) \end{aligned}$$

Which is a closed set in $\operatorname{Spec}(S)$, and hence the map $\varphi^{\#}$ is continuous.

Problem 2 (5 points). Let R be a ring and let $f_1, f_2, \ldots, f_t \in R$ be non-zero elements such that $\{D(f_i)\}_{i=1}^t$ form an open cover of $\operatorname{Spec}(R)$. Carefully prove that R is reduced if and only if $R\left[\frac{1}{f_i}\right]$ is reduced for all i.

Solution.

$(\implies)$

Assume R is reduced. That is $\operatorname{Nil}(R) = \langle 0 \rangle$. And by definition for $f_i \neq 0$,

$$R\left[\frac{1}{f_i}\right] = \left\{\frac{r}{f_i^n} \mid r \in R,\, n \geq 0\right\}$$

Now assume $\left(r/f_i^n\right)^m = 0$ for some $m > 0$. Then, we have

$$\begin{aligned} (r/f_i^n)^m &= \frac{0}{1} \\ f_i^k(r^m \cdot 1 - 0 \cdot f_i^{nm}) &= 0 \\ f_i^k r^m &= 0 \\ \implies (f_i^k)^{m-1}(f_i^k r^m) = (f_i^k r)^m &= 0 \end{aligned}$$

But since R is reduced, this implies that $f_i^k r = 0$, and thus $r/f_i^n = 0 \in R\left[\frac{1}{f_i}\right]$. Hence $R\left[\frac{1}{f_i}\right]$ has no non-zero nilpotents and is reduced.

$(\impliedby)$

Assume $R\left[\frac{1}{f_i}\right]$ is reduced for all i. Let $r \in R$ such that $r \neq 0$, $r^m = 0$. Then, in each $R\left[\frac{1}{f_i}\right]$ we have,

$$\left(\frac{r}{1}\right)^m = \left(\frac{0}{1}\right)$$

But since each $R\left[\frac{1}{f_i}\right]$ is reduced this implies that $\frac{r}{1} = 0 \in R\left[\frac{1}{f_i}\right]$. Then, there exists $f_i^{k_i}$ for each i such that,

$$\begin{aligned} f_i^{k_i}(r - 0 \cdot 1) &= 0 \\ f_i^{k_i} r &= 0 \in R \end{aligned}$$

Now we consider $\operatorname{Ann}(r)$. For all i, $f_i^{k_i} \in \operatorname{Ann}(r)$. We claim that $\operatorname{Ann}(r)$ is not contained in any prime ideal in R.

Suppose that it were, then $\operatorname{Ann}(r) \subseteq P$ for some prime ideal P and since prime ideals are radical this implies that $f_i \in P$ for all i. But since by assumption $\{D(f_i)\}_{i=1}^t$ form an open cover of $\mathrm{Spec}(R)$ at least one $f_i \notin P$. This leads to a contradiction and hence $\mathrm{Ann}(r) \notin \mathrm{Spec}(R)$.

But since every proper ideal is contained in some maximal ideal (and thus some prime ideal) this implies that $\mathrm{Ann}(r)$ is not proper. Then we must have $\mathrm{Ann}(r) = R$, i.e. $1 \in \mathrm{Ann}(r)$ and hence $r = 0$. Thus, R is reudced.